

Brian Munene Mwirigi

Software Engineer | AI/ML Engineer | Entrepreneur

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PROFESSIONAL SUMMARY

Second-year Computer Science student at Strathmore University with a strong track record of shipping production-grade AI/ML systems and developer tools. Co-founder of Corvux, a sovereign AI infrastructure company; AI/ML Lead at CropChain (UN PRME semi-finalist). Published ML researcher with a Zenodo paper on computational consciousness. Builder of Chrome extensions and CLI tools with 400+ active users. Experienced across the full stack from transformer architecture research to Next.js/React Native front ends and FastAPI back ends.

EXPERIENCE

AI/ML Lead | [CropChain \(UN PRME Innovation Studio\)](#) Sep 2025 - Present | Nairobi, Kenya

- Trained XGBoost credit-scoring model on 4,563 records (Zindi Financial Inclusion dataset) achieving AUC-ROC of 0.7031.
- Built production FastAPI + PostgreSQL backend with Sentinel-2/CDSE satellite imagery pipeline and Open-Meteo weather integration.
- Integrated M-Pesa transaction analysis for alternative credit signals; built React Native Android app with 'Organic Intelligence' UI.
- Conducted 13 stakeholder interviews; advanced CropChain to PRME Innovation Studio semi-finals (pitched May 2026).

Founder | [Breon Studio](#) Jan 2026 - Present | Nairobi, Kenya

- Runs a web development and design studio delivering production websites for clients including Metumi Paints Centre
- Full-stack delivery across Next.js, React, Tailwind CSS, and M-Pesa/Africa's Talking payment integrations.

OPEN SOURCE & PRODUCTS

AI Slop Blocker | [Chrome Extension](#) 2026

- 415+ active users; local Naive Bayes ML classifier at 94.8% accuracy to detect AI-generated content in-browser.
- Monetized via Lemon Squeezy (\$12/month or \$120 lifetime); built ML pipeline entirely client-side to preserve privacy.

CostHQ (formerly codesession-cli) | [Open-Source CLI Tool](#) 2026

- AI cost-tracking CLI for developers; open-core freemium model with pro tier gating (proxyCaching, advancedFirewall, multiProject).
- Organic npm traction alongside sister packages: aitoken-cli, runbook-cli, devmem-cli, apisnap-cli.

MeridianArc | [Desktop App](#) 2026

- Cross-platform Pomodoro productivity app built with Tauri + React + TypeScript; scaffolded and shipped to GitHub.

RESEARCH & PUBLICATIONS

Computational Consciousness Simulation | [Zenodo \(DOI: 10.5281/zenodo.18940956\)](#) 2026

- Published multi-agent ML research finding that network topology — not learning — determines a network's ability to predict its own behavior.
- Identified phase transition between N=324 and N=576 agents; measured autonomy-predictability anti-correlation ($r = -0.71$).

ConstKV / Engram - KV Cache Compression | [Independent Research](#)

2026

- Designed novel transformer KV-cache compression architecture: SSM-based encoder compressing token sequences into a fixed-size latent state, with query-conditioned MLP reconstructor.
- Smoke test passed with stable loss (0.7943); compression ratios from $3,072\times$ ($T=128$) to $98,304\times$ ($T=4,096$).

Panacea - Vultr-Native Zero-Trust Security Agent for Hospital Networks | [RAISE Summit Hackathon \(Paris\)](#)

2026

- Built autonomous multi-step AI agent (Agent Orchestration lead) that reads medical device manuals, cross-checks CVE databases and enforces live zero-trust firewall policy on Vultr Cloud Firewall
- Designed the agentic loop, per-rule citation trail, explainable confidence scoring, and audit-log system; shipped on Vultr Serverless Inference (Nemotron) and Vultr Vector Store for a HIPAA-aligned, zero-egress pitch-
- Submitted to RAISE Summit Hackathon, Paris (July 2026).

EDUCATION

BSc Computer Science | [Strathmore University](#)

2024 - Present | Nairobi, Kenya

- Second year. Relevant coursework: Algorithms, Data Structures, Machine Learning, Database Systems, Software Engineering.

Software Engineering Certificate | [Moringa School \(Flatiron School Partnership\)](#)

2025 | Nairobi, Kenya

- Full-stack web development and software engineering fundamentals; completed prior to Strathmore enrollment.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Rust (learning), Bash

AI/ML: PyTorch, Transformers (HuggingFace), XGBoost, Scikit-learn, LangChain, CrewAI, Langfuse, RAGAS, Presidio

LLM Infra: LlamaFarm, Ollama, vLLM, LLM Guard, Vanna.ai, Graphiti, Llama 3 / Mistral / Qwen / DeepSeek / Gemma 4

Backend: FastAPI, Node.js, PostgreSQL, REST APIs, Africa's Talking, M-Pesa Daraja API

Frontend: React, Next.js, React Native, Tailwind CSS, Tauri

DevOps: Docker, GitHub Actions, Railway, Linux/WSL, Git

Other: Chrome Extension APIs, npm publishing, Lemon Squeezy monetization, Sentinel-2/CDSE satellite data

AWARDS & COMPETITIVE PROGRAMS

- UN PRME Innovation Studio Semi-Finalist CropChain · May 2026